

$$\omega = \sqrt{\omega_0^2 - \lambda^2}$$

$$\omega^2 = \omega_0^2 - \lambda^2$$

$$\omega_0^2 = \omega^2 + \lambda^2$$

$$\omega_0 = \sqrt{\omega^2 + \lambda^2}$$

$$\Delta f = \sqrt{\left(\frac{\partial f}{\partial x}\Delta x\right)^2 + \left(\frac{\partial f}{\partial y}\Delta y\right)^2}$$

$$\Delta \omega_0 = \sqrt{\left(\frac{\partial \omega_0}{\partial \omega}\Delta \omega\right)^2 + \left(\frac{\partial \omega_0}{\partial \lambda}\Delta \lambda\right)^2}$$

$$\frac{\partial \omega_0}{\partial \omega} = \frac{\partial}{\partial \omega} \left(\sqrt{\omega^2 + \lambda^2} \right) = \frac{\omega}{\sqrt{\omega^2 + \lambda^2}} = \frac{\omega}{\omega_0}$$

$$\frac{\partial \omega_0}{\partial \lambda} = \frac{\partial}{\partial \lambda} \left(\sqrt{\omega^2 + \lambda^2} \right) = \frac{\lambda}{\sqrt{\omega^2 + \lambda^2}} = \frac{\lambda}{\omega_0}$$

$$\Delta \omega_0 = \sqrt{\left(\frac{\omega}{\omega_0}\Delta \omega\right)^2 + \left(\frac{\lambda}{\omega_0}\Delta \lambda\right)^2}$$

$$\Delta \omega_0 = \frac{1}{\omega_0} \sqrt{(\omega \Delta \omega)^2 + (\lambda \Delta \lambda)^2}$$

$$\Delta \omega_0 = \frac{1}{\omega_0} \sqrt{(\omega \Delta \omega)^2 + (\lambda \Delta \lambda)^2}$$